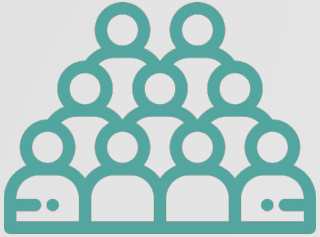
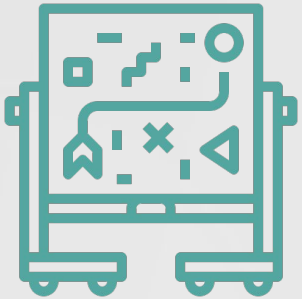
COS Office Hour Queue

Kelly Petrino | COS IW04

Motivation



Office Hours for COS classes are overcrowded



Current queuing system is inefficient

TAs asked same questions over and over again

Goal

Problem: Office Hours as they currently exist do not meet the needs of students and instructors.

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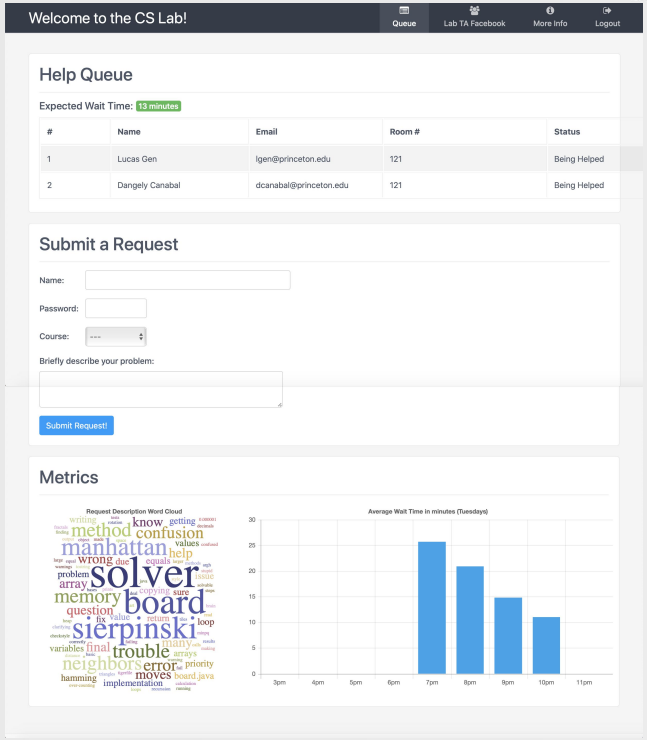
Create an **online platform** that allows students to join an **online queue** for Office Hours to make the experience **more efficient and simplistic** for both students and TAs.

Related Work

COS Lab TA Queue

- Inspiration for this project
- Online queue with estimated wait times and problem descriptions
- Can be improved upon → **Link students with similar questions to work together**



Related Work

"Collaboration may be a defining feature of competent and optimal practice, and the failure to collaborate may be indicative of negligence and malpractice."

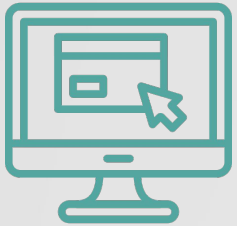
The logic of collaboration in education and the human services
- Hal A Lawson (2004)

Small group collaborative learning (SGCL) shows an increase in individual critical thinking skills for science and math learning.

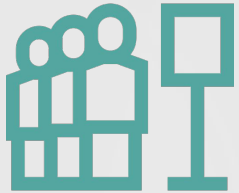
Critical thinking in higher education: The influence of teaching styles and peer collaboration on science and math learning - I.J. Quitadamo (2002)

Approach

There does not currently exist an efficient way for large groups of students to have their questions answered in COS office hours.



Develop a web application with different interfaces for students and instructors

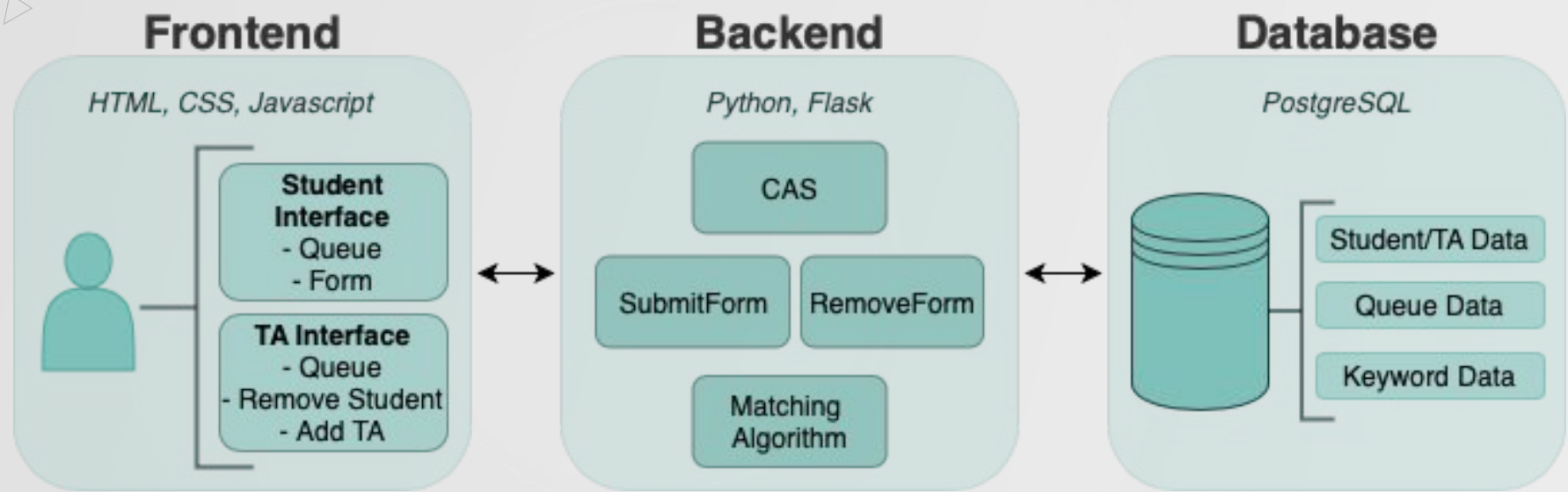


Create an online queue with an estimated wait time for students attending OH

Design a matching program to link students with similar problems to work together

Implementation

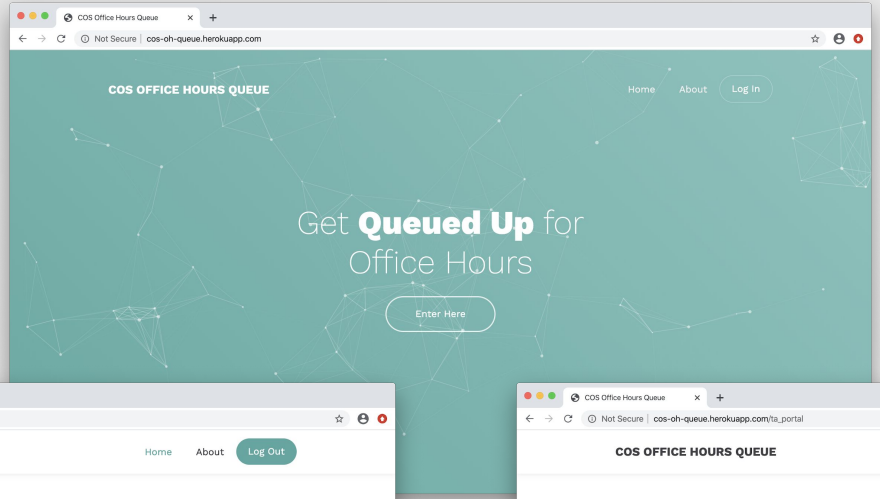
3-tier Web Application hosted on Heroku

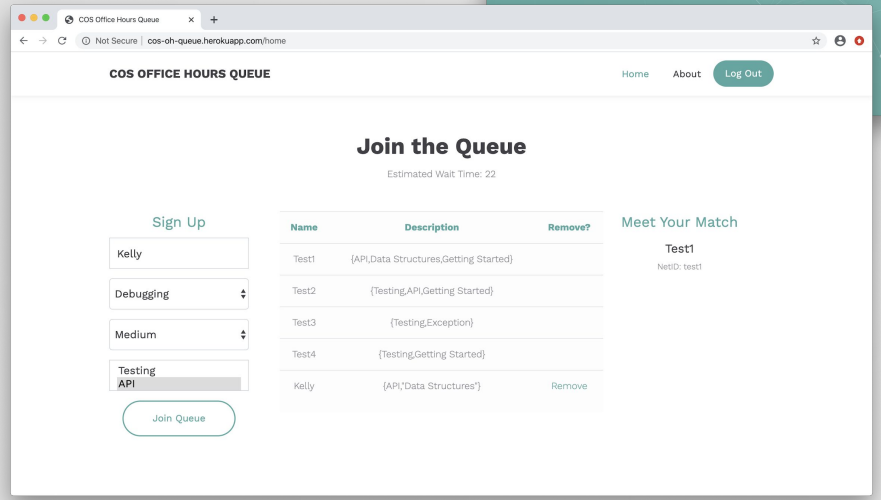


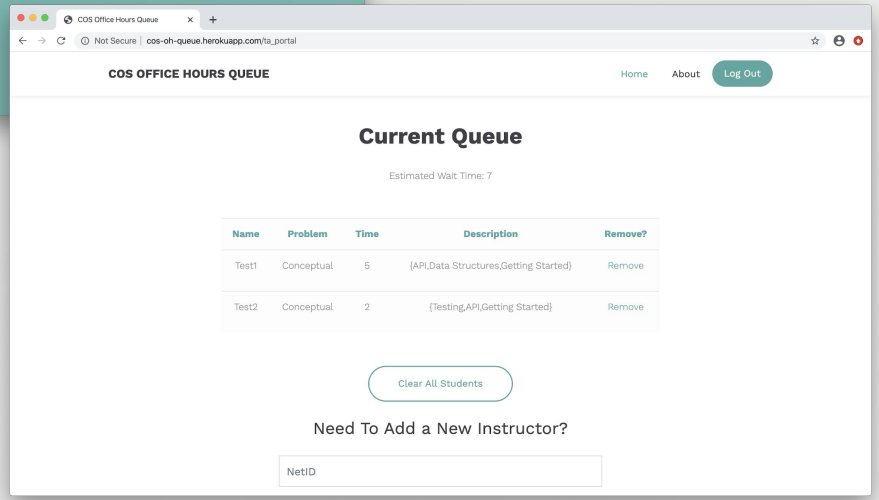
Implementation

Students

Instructors



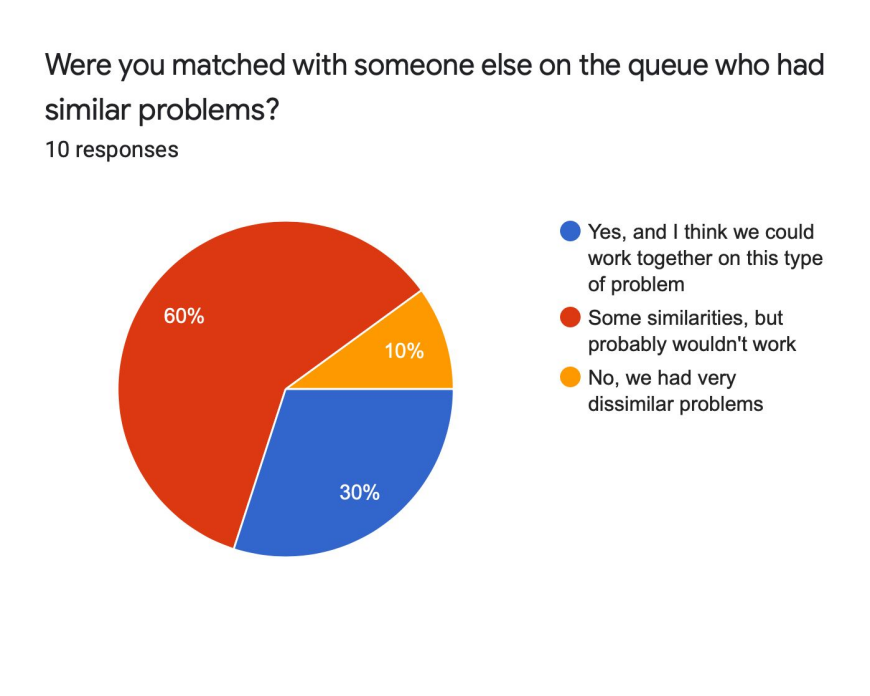




Evaluation

Setup: Gathered 10 Princeton students to test the application at once to **simulate an Office Hours Session**.

Procedure: Had each student utilize the application as though they were actually attending OH and then fill out a **survey about the applications performance and usefulness**.



Conclusion

The COS Office Hours Queue accomplishes its goal of creating an online platform that allows students to join a virtual queue to receive help and be matched with other students to tackle similar problems while they wait.

Students can join the queue and see their expected wait time

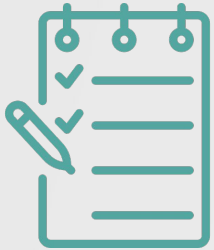
Students can describe their problems to be matched with others

TAs can manage the queue and avoid repeat questions

Future Work

Remaining Steps:

- Finish evaluating and debugging the application
- Implement simple changes based on feedback from evaluation



Long Term:

- Expand the application to work for multiple differents classes
- Implement large scale changes based on feedback (Zoom functionality, being matched in larger groups, etc.)

Acknowledgements

A big thank you to my advisor for this project, **Dr. Robert Fish**, along with my TA, **Carlo Rosati**, and the entire **IW04 class** for all of your input and guidance over the semester.

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THANK YOU